

CAPSTONE PROJECT ASSIGNMENT SHEET

Project Sentinel: Hazardous Asteroid Prediction AI

1. Project Overview

At NASA's Planetary Defense Coordination Office, thousands of Near Earth Objects (NEOs) are continuously monitored to assess potential risks to Earth. While the majority of these asteroids and comets are harmless, a small percentage are classified as potentially hazardous due to their size, velocity, proximity to Earth, and orbital characteristics.

As a Data Science Intern, your objective is to develop an intelligent Machine Learning system capable of automatically predicting whether a Near Earth Object should be classified as Hazardous or Non-Hazardous. By leveraging historical NASA observations, the proposed solution should assist researchers in analyzing large astronomical datasets more efficiently and support data-driven risk assessment.

2. Data Architecture & Core Challenges

You have been provided with a historical NASA Near Earth Object dataset containing physical and orbital characteristics of thousands of asteroids.

The dataset contains the following key attributes:

- Absolute Magnitude

- Estimated Diameter Minimum
- Estimated Diameter Maximum
- Relative Velocity
- Miss Distance
- Orbital Parameters
- Hazardous Label (Target Variable)

To complete this assignment successfully, your implementation must address the following challenges:

1. Multi-Factor Risk Assessment Challenge

A single feature cannot determine whether an asteroid is hazardous.

For example, a large asteroid may pass safely at a significant distance from Earth, while a smaller object traveling at a high velocity and approaching more closely may require greater attention.

Your model must learn relationships between multiple physical and orbital characteristics simultaneously.

2. Feature Relationship Challenge

The interaction between asteroid size, velocity, distance, and orbital properties is often complex and non-linear.

Simple rule-based systems may fail to capture these relationships.

Your model must identify hidden patterns within the data that distinguish hazardous objects from non-hazardous objects.

3. Class Imbalance Challenge

Hazardous asteroids represent only a small percentage of the overall dataset.

As a result, a model that simply predicts every object as non-hazardous may still achieve misleadingly high accuracy.

You must implement strategies that properly handle class imbalance and prioritize the identification of hazardous objects.

4. Explainable AI Challenge

Scientists require justification for model predictions.

You must identify which features contribute most strongly to hazard classification and provide visual explanations supporting the model's decision-making process.

3. Phase-by-Phase Milestones

Your implementation must progress through the following phases within a single Jupyter Notebook.

Phase 1: Data Preparation & Preprocessing

- Load the NASA NEO dataset.
- Handle missing values.
- Remove unnecessary attributes.
- Encode categorical variables if required.
- Normalize or scale numerical features where appropriate.

Phase 2: Exploratory Data Analysis

- Analyze class distributions.
- Visualize relationships between important features.
- Identify patterns associated with hazardous objects.
- Generate summary statistics and visual insights.

Phase 3: Feature Engineering & Selection

- Select the most relevant predictive features.
- Remove redundant attributes.
- Evaluate feature correlations.
- Prepare the final feature matrix for modeling.

Phase 4: Model Development

- Split the dataset into training and testing subsets.
- Train one or more supervised classification algorithms.
- Optimize model performance using suitable hyperparameters.
- Compare model performance across different approaches.

Phase 5: Explainability & Visualization

- Generate feature importance visualizations.
- Interpret the model's predictions.

- Identify the strongest indicators of hazardous asteroids.
- Provide a brief scientific interpretation of the findings.

Phase 6: Deployment

- Save the final trained model.
 - Develop a simple user interface capable of accepting asteroid characteristics and generating predictions.
 - Display whether the object is Hazardous or Non-Hazardous.
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4. Final Problem Statement

NASA possesses historical data describing Near Earth Objects using features such as Absolute Magnitude, Estimated Diameter, Relative Velocity, Miss Distance, and Orbital Parameters.

However, manually analyzing thousands of observations to identify potentially hazardous objects is both difficult and time-consuming.

Objective:

Develop a supervised machine learning classification system capable of predicting whether a Near Earth Object is Hazardous or Non-Hazardous based on its physical and orbital characteristics.

Input:

- Absolute Magnitude
- Estimated Diameter
- Relative Velocity
- Miss Distance
- Orbital Parameters

Output:

- Hazardous
- Non-Hazardous

Machine Learning Task:

Supervised Binary Classification

5. Final Submission Standards & Evaluation

Deliverable Constraint:

Submit exactly one Jupyter Notebook named:

hazardous_asteroid_prediction_final.ipynb

The notebook must execute sequentially from a clean runtime environment without errors.

Evaluation Criteria:

- Data Preprocessing & Cleaning (20%)
- Exploratory Data Analysis (20%)
- Feature Engineering & Selection (20%)
- Model Performance & Evaluation (20%)
- Explainability, Visualization & Deployment (20%)